

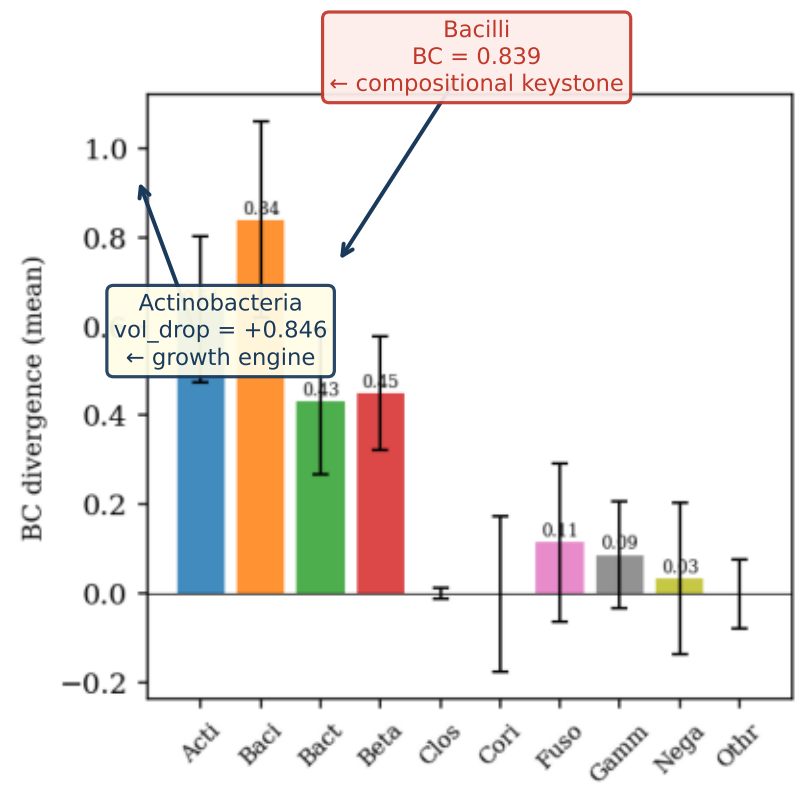
Guild-Level Dynamical Analysis

Keystone · Network Centrality · Relapse Dynamics · Prevention · Phase Diagram

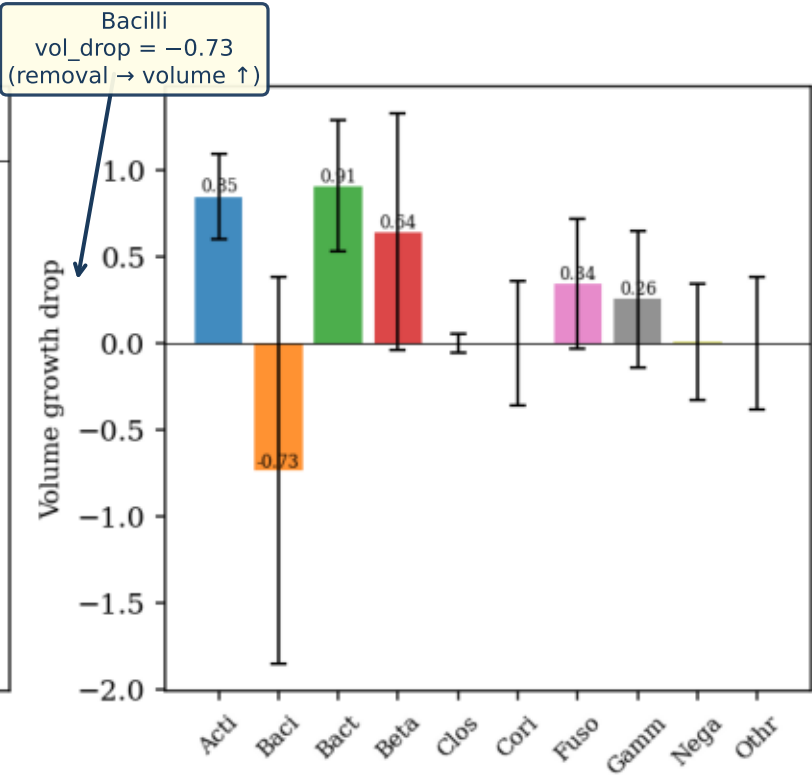
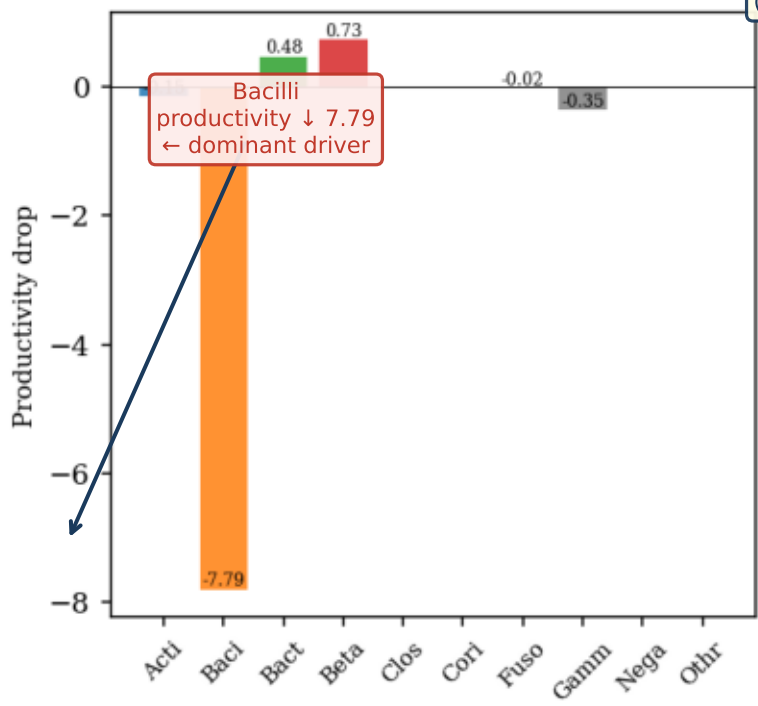
Dieckow 2024 · 10-guild gLV · $N = 10$ patients · Nishioka (2026)

Guild-Level Keystone Analysis: Knockout BC Divergence

Remove one guild → simulate gLV 21 d → measure BC shift, productivity drop, volume change



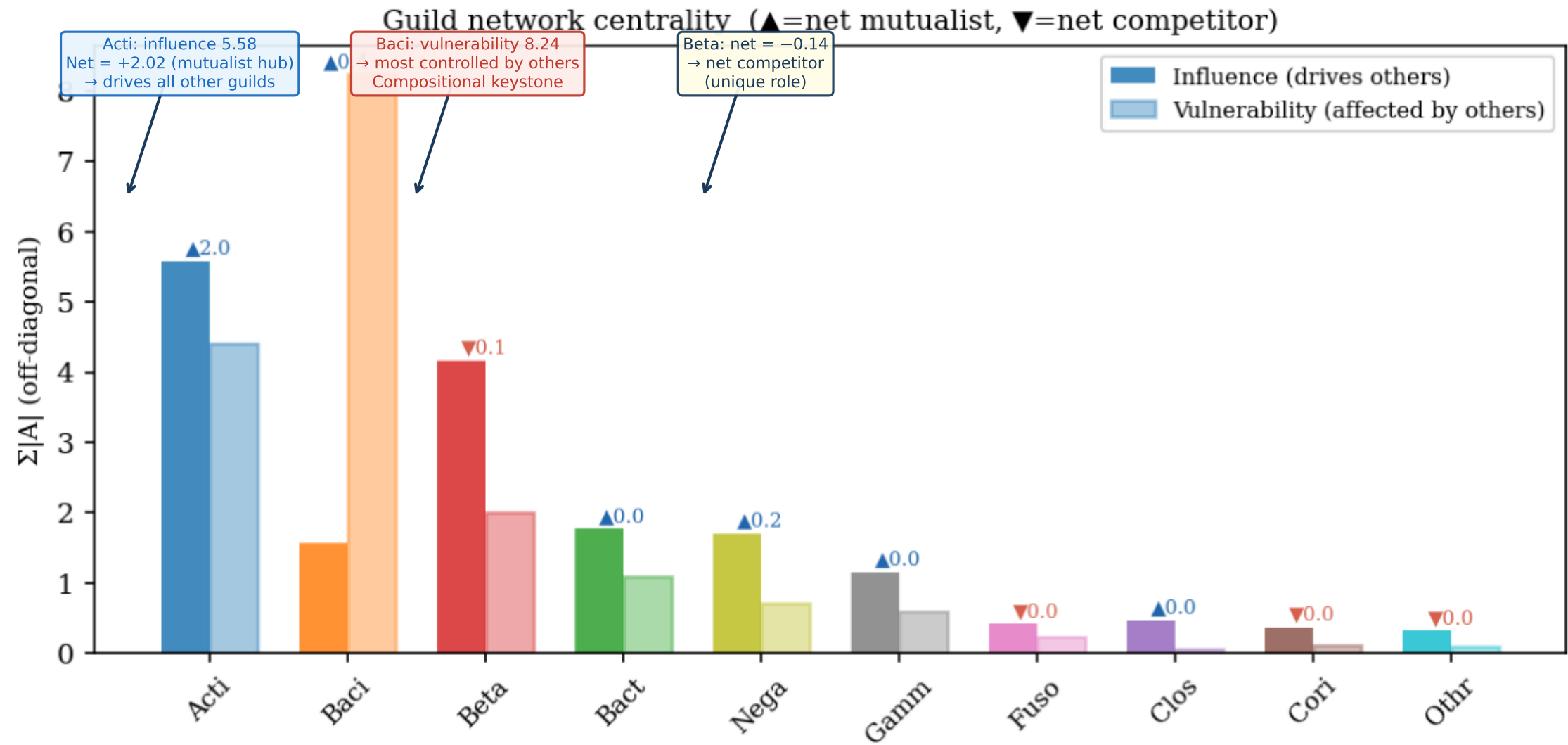
10-guild knockout effects (Dieckow gLV)



Bacilli: highest BC divergence (0.839) — compositional keystone. Removal increases total volume → competitive suppressor. | Actinobacteria: vol_drop = +0.846 → growth engine. Strongest single interaction: Acti → Baci A = +3.43.

Guild Network Centrality: Influence, Vulnerability, Net Role

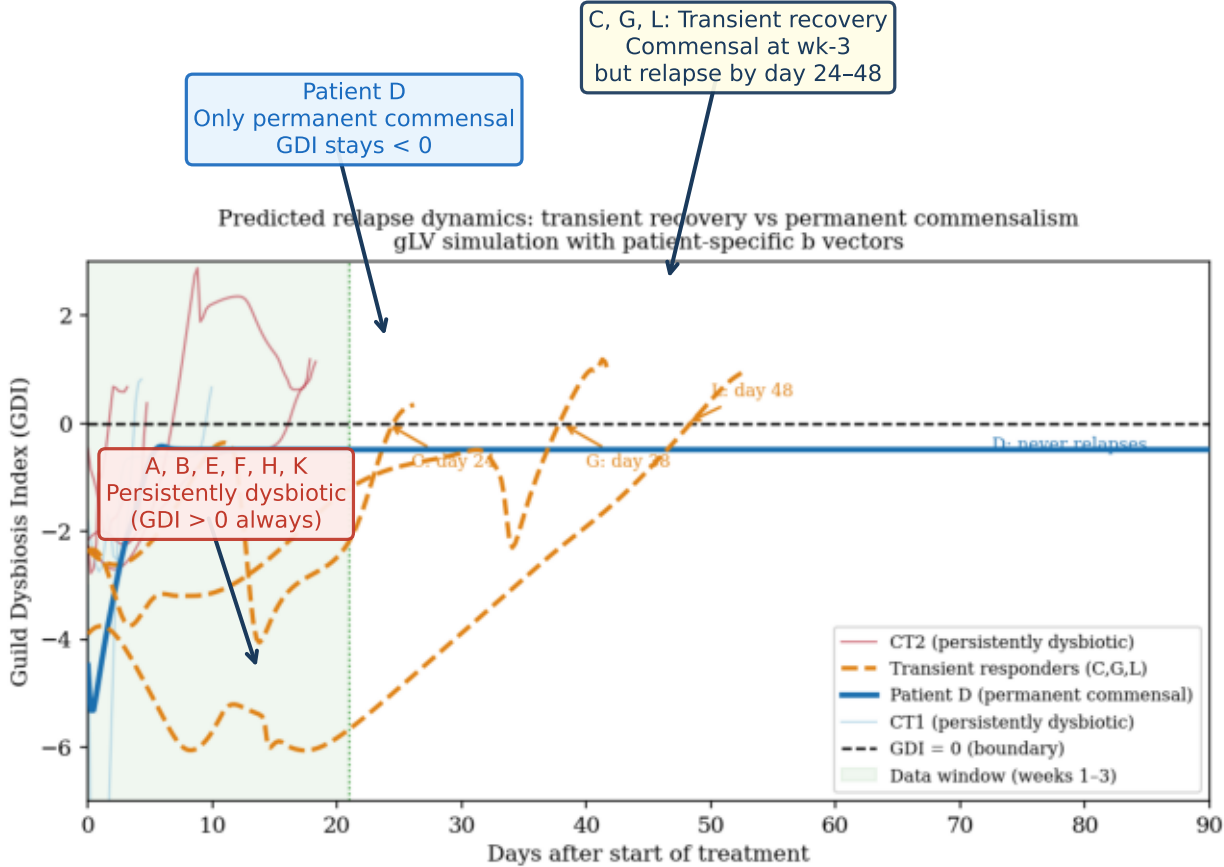
Influence = $\sum |A_{ji}|$ (how much guild i drives others) | Vulnerability = $\sum |A_{ij}|$ (how much others drive guild i)



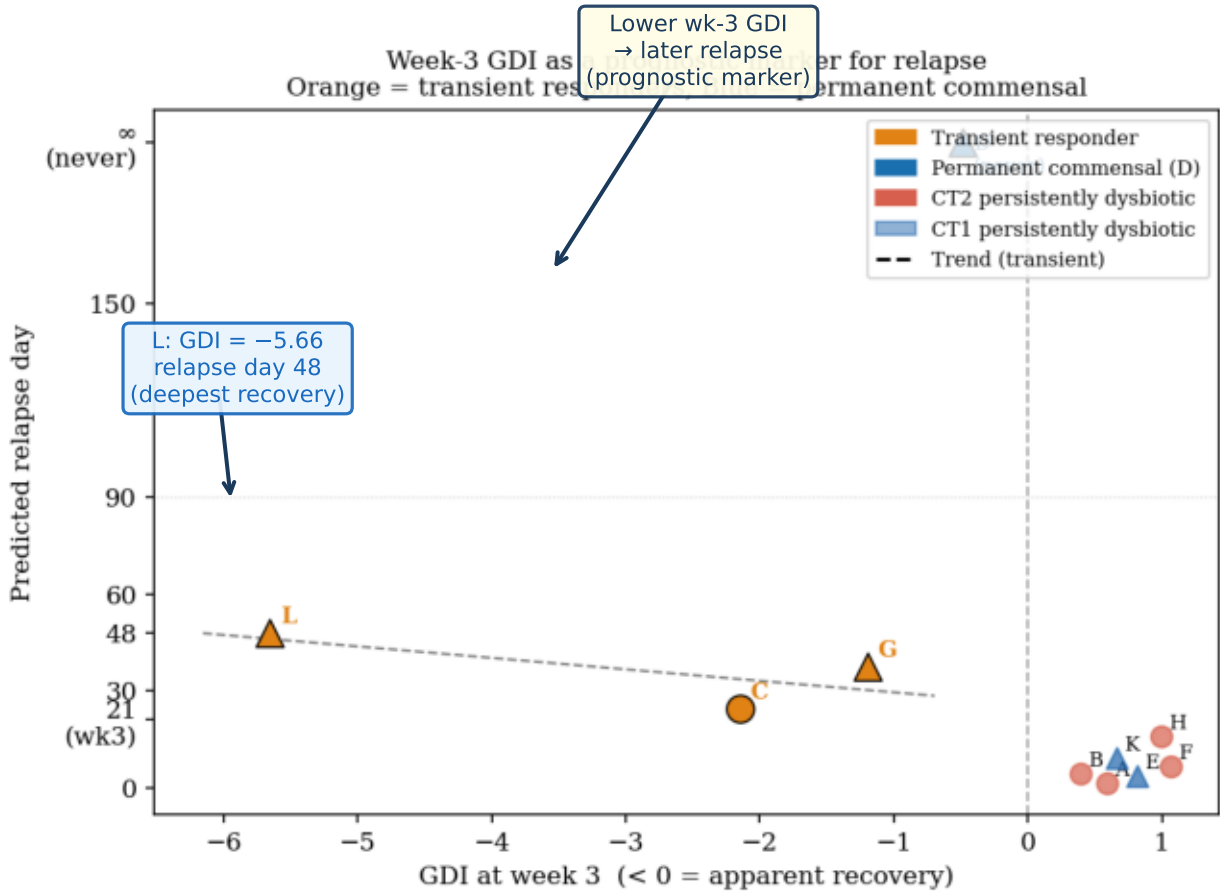
Actinobacteria: influence = 5.58 (highest), net = +2.02 → mutualist hub. Bacilli: vulnerability = 8.24 → most regulated by others. Betaproteobacteria: net = −0.14 → net competitor. A matrix encodes dysbiotic attractor via Acti-Baci-Beta triad.

Predicted Relapse Dynamics: Three Patient Archetypes

gLV simulation t = 150 d with patient-specific b vectors | $GDI = \log(\phi_{dys}) - \log(\phi_{com})$



(a) GDI trajectories to 90 d

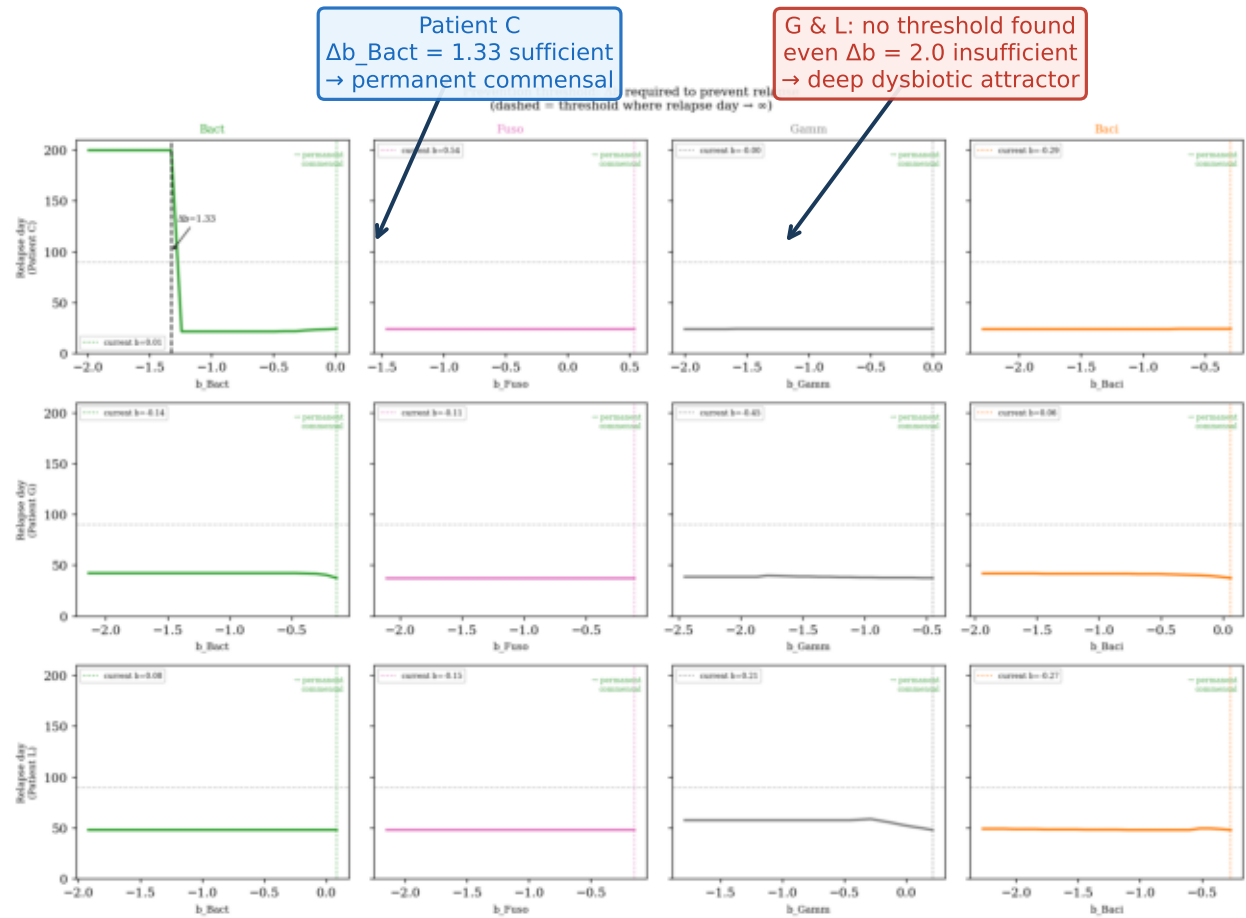


(b) Week-3 GDI as prognostic marker

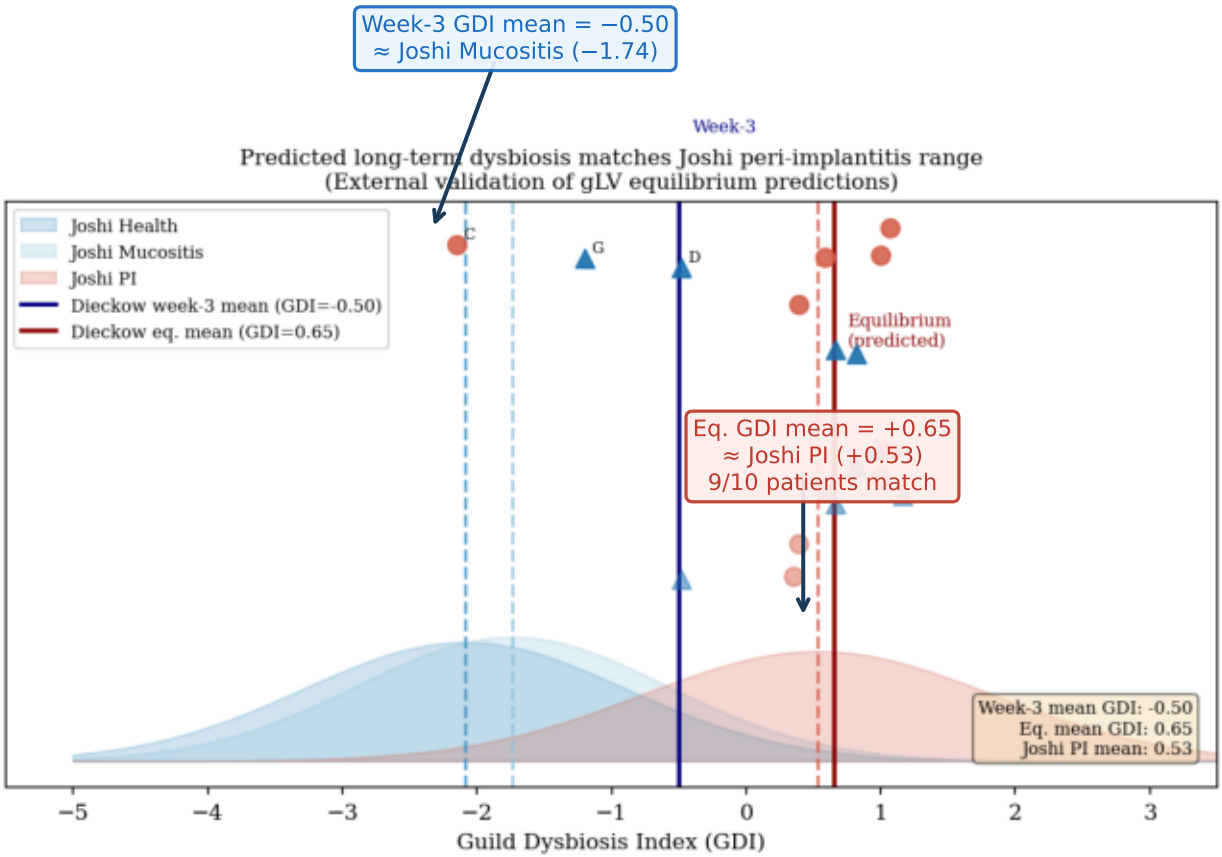
① Persistent dysbiotic (A,B,E,F,H,K): $GDI > 0$ throughout. ② Transient responders (C→24d, G→38d, L→48d): commensal at wk-3, dysbiotic by day 50. ③ Permanent commensal (D only): stable $GDI < 0$. → Week-3 GDI as prognostic marker: lower = longer remission.

Prevention Threshold & External Validation (Joshi 2025)

Min. Δb_i to prevent relapse (single-guild sweep) | Model eq. GDI vs peri-implantitis reference distribution



(a) Required Δb per guild for prevention

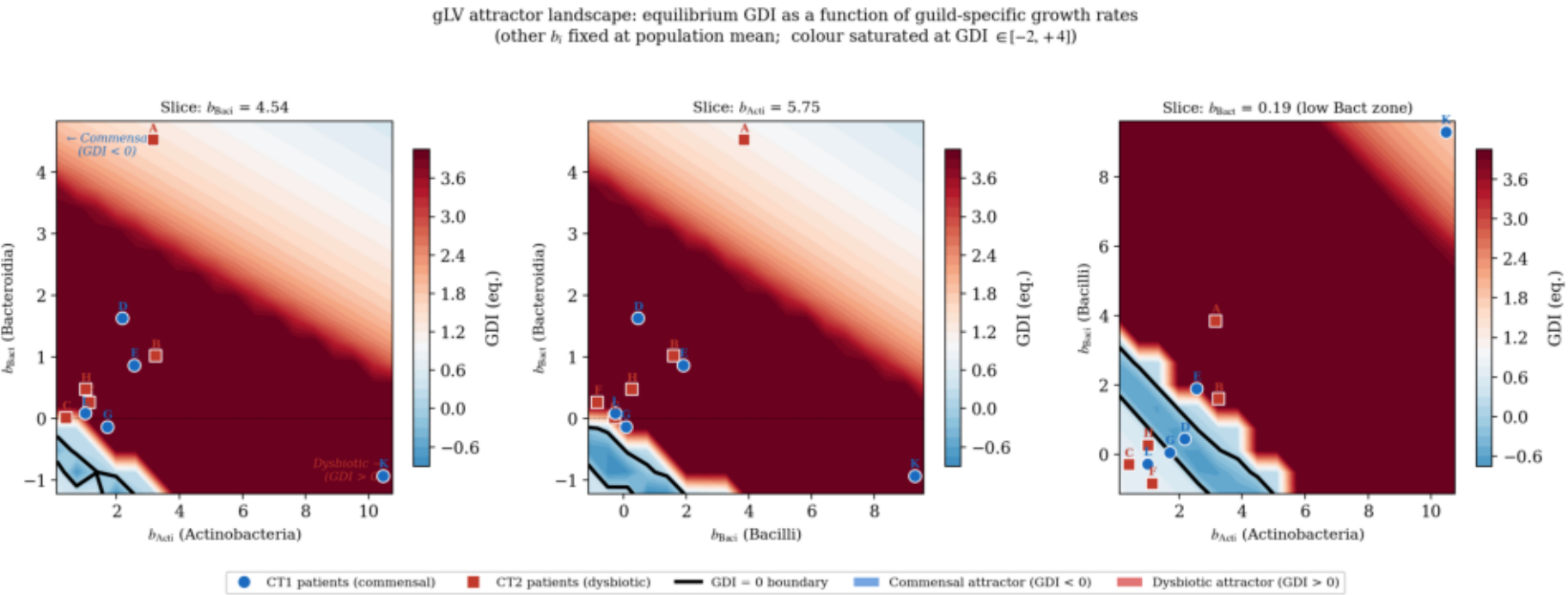


(b) External validation: Joshi 2025 peri-implant cohort

Prevention: only Patient C convertible by single-guild intervention ($\Delta b_{Bact} = 1.33$). G & L require multi-guild change $\rightarrow$ deep attractor. | Joshi validation: predicted eq. GDI = +0.65 $\approx$ Joshi PI mean +0.53 (9/10 patients; difference $< 0.12 <$ Joshi SD = 1.30).

3D Attractor Landscape: GDI = 0 Boundary in Growth-Rate Space

18³ grid, Acti × Baci × Bact, t = 80 d | Blue = commensal | Red = dysbiotic | Black = GDI = 0 boundary



Key findings

98% of b-space

is dysbiotic

(GDI > 0)

Commensal zone:

$b_Bact < \sim 1.5$

only 2% of space

b_Bact (Bacteroidia)

= dominant axis

boundary at ~ 1.5

CT1 (○): near boundary

CT2 patient A:

$b_Bact=4.5 \rightarrow$ deep dys.

$\rightarrow$ structural attractor

98% of parameter space is dysbiotic (GDI > 0). Commensal region confined to $b_Bact < \sim 1.5$ — Bacteroidia growth rate is the dominant control variable. CT1 patients with low b_Bact cluster near boundary; CT2 patient A ($b_Bact = 4.5$) is deep in dysbiotic zone. $\rightarrow$ Structural attractor robustness — not initial-condition sensitive.